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THE EFFECTS OF ACUTE SOCIAL MEDIA EXPOSURE ON BODY DISSATISFACTION AND EATING BEHAVIOR OF MALE AND FEMALE STUDENTS

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Introduction: Experimental research has examined the effect of social media on body dissatisfaction, but little attention has been given to the impact on eating behavior. Across two studies, we tested the causal relationship between acute social media use, body dissatisfaction and eating behavior. **Methods:** In Study 1, 80 female participants (age: M = 20.75 years; BMI: M = 21.3) viewed their own Facebook account or a news website before completing a portion size selection

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task. In Study 2 (pre-registered), 148 participants (81 females; age: M = 20.92 years; BMI: M = 22.19), viewed their own or an experimenter-curated Instagram profile before consuming potato chips ad-libitum. **Results:** In Study 1, body dissatisfaction was significantly higher in participants who viewed Facebook (compared to a news website) and significantly mediated the effect of condition on portion size selection. In Study 2, regardless of condition females reported significantly smaller ideal body sizes and higher body dissatisfaction compared to males. When females reported a thinner body size as ideal, females in the personal Instagram consumed significantly less potato chips than those in the control Instagram condition. When males reported a smaller actual body size, males in the personal Instagram condition consumed significantly less potato chips than males in the control Instagram condition. **Discussion:** Findings suggest brief bouts of social media use may produce immediate disruptions to body image of both sexes, and restrained snack intake of females endorsing thin ideals.

Keywords: Body Image, Body Dissatisfaction, Social Media Use, Eating Behaviors, Portion Size

INTRODUCTION

Prior research has suggested the negative impact of social media on body image (Fardouly & Vartanian, 2016; Saiphoo & Vahedi, 2019). This has included reports of internal research from Meta (formerly Facebook) showing that Instagram (a photo sharing platform that is a core Meta product) is harmful to mental health and body image due to social comparisons that influence how women view and describe themselves (Wells et al., 2021). The report highlighted the platform's influence on vulnerable individuals, such as younger users and how individuals with mental health conditions report feeling worse after using the platform. Broader reviews also show greater body image concerns have

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been linked to social media use, which are exacerbated by appearance-focused social media especially among younger age users (Saiphoo & Vahedi, 2019).

A wealth of correlational studies emphasize the relationship between social media and body dissatisfaction (Bennett et al., 2020; Griffiths et al., 2018; Lee & Lee, 2021; McLean et al., 2015; Modica, 2020; Saunders & Eaton, 2018; Stronge et al., 2015). In particular, image based social media (e.g., Facebook and Instagram) has been associated with poor body image and negative affect. For example, experimental studies suggest that viewing images of thin and attractive individuals (commonly seen on social media sites) can lead to increased body dissatisfaction and negative mood (Fardouly & Holland, 2018; Livingston et al., 2020; Tiggemann & Anderberg, 2020; Tiggemann et al., 2018). Tiggemann and colleagues (2018) found individuals reported increased body and facial dissatisfaction if they had viewed thin-ideal images compared to average weight images (presented in an Instagram format), and Tiggemann and Anderberg (2020) found greater body dissatisfaction and lesser body appreciation reported by individuals who only viewed the ideal images from 'Instagram vs reality' posts. It is likely that upward comparisons (i.e., comparing oneself with those deemed more superior) may be a cause. For example, Hogue and Mills (2019) found increased negative body image in females after browsing and commenting on a Facebook profile for an attractive female peer compared to females who completed the same task targeting a female family member. Levels of comparison (both high and low) have shown to be similar across genders (Gomez et al., 2021). Consequently, the current studies focus on social media that is primarily image based.

Body dissatisfaction is a key maintenance factor for eating disorders (Stice, 2002). Prior research has also shown associations between social media use, body dissatisfaction, and disordered eating behaviors (Holland & Tiggemann, 2016; Saiphoo & Vahedi, 2019). A range of social media indices (i.e., frequency of use, time spent on social media, number of friends) are associated with body image concerns and disordered eating behaviors according to several reviews (Holland & Tiggemann, 2016; Ioannidis et al., 2021; Rounsefell et al., 2020; Ryding & Kuss, 2020). Examples of specific associations include (a) higher negative social media engagement and greater body dissatisfaction and

disordered food choices (Rounsefell et al., 2020), (b) increased thoughts of dieting in individuals who make upward comparisons (comparing oneself to others better off (Gibbons, 1986) after comparing themselves to social media images (Fardouly et al., 2017) and (c) increased feelings of hunger and distraction from pursuing healthy eating goals after viewing images of food (Vaterlaus et al., 2015). Despite a plethora of association studies and experimental studies with disordered eating behavior as the outcome (see reviews Holland & Tiggemann, 2016; Ioannidis et al., 2021), to date there are limited experimental studies that examine the potential causal impact of exposure to social media on subsequent eating behaviors (i.e., food intake and portion size). Here, we advanced previous findings by experimentally examining the impact of social media on body image and eating behaviors using portion size selection and snack intake tasks.

There are several possible mechanisms in which social media may affect subsequent portion selection patterns and eating behavior. Firstly, individuals may alter what they eat based on their perceptions of what social media users do and should eat. For example, Hawkins and colleagues (2020) found fruit and vegetable intake was significantly predicted by perception of Facebook users' intake while high energy dense snack consumption was predicted by perceptions of what users should eat. Secondly, exposure to peers' meticulously curated social media posts that largely present their desirable experiences, traits (e.g., best looking pictures of oneself), and affirmations of support from others may produce feelings of relative disadvantage among viewers. Indeed, the concept of fear of missing out (FOMO) which is defined as the state of anxiety over being absent from rewarding experiences that others are having, has been associated with poor well-being and depression (Hunt et al., 2018). Perceptions of falling behind others can increase an individual's desire to select and consume larger portion sizes (Sim, Lim, Forde, & Cheon, 2018). Therefore, feeling like one is missing out and associated feelings linked to emotional eating may lead to increase food consumption (Cheon et al., 2018; Cheon & Hong, 2017; Sim, Lim, Leow, & Cheon, 2018).

Yet, it is also possible that social media use may subsequently reduce or restrain food intake. Disturbances to body image and eating behavior following social media use may be a product of specific types of social media content, such as selectively curated

images that make peers seem especially physically attractive (e.g., enhanced or modified selfies; Cohen et al., 2018). Although broader upward social comparison with others on social/economic resources may stimulate appetite (Cheon & Hong, 2017; Sim, Lim, Forde, et al., 2018), it is also possible that domain specific upward comparisons on physical appearance may selectively lead to intentions to restrict food intake to reduce perceived discrepancies in body size/appearance with others.

CURRENT STUDIES

The current studies aimed to take initial steps in establishing a causal link between social media, body dissatisfaction, and eating behavior. The overarching aim was to experimentally examine the influence of social media on body image (aim 1) and eating behaviors (aim 2), specifically portion size and food intake. We first predicted that a brief exposure to naturalistic social media usage (i.e., browsing one's account for less than 10 minutes) will be associated with greater body dissatisfaction (H1) and desire to consume less food (H2), in other words reduced portion size selection and snack intake, compared to using other types of digital media (e.g., browsing daily news). We also examined factors that may mediate the relationship between social media exposure and eating behaviors (aim 3). We hypothesized that increased body dissatisfaction would significantly mediate the relationship between social media exposure and eating behavior (H3). Individuals that viewed social media were hypothesized to experience an increase in body dissatisfaction and in turn, reduce their portion size selection or food intake. Study 1 and 2 examined these behaviors in the context of Facebook and Instagram use on the influence of portion size and actual food intake, respectively.

Individuals often use Facebook and Instagram by scrolling through feeds compiled of the latest posts. In these feeds individuals are exposed to a range of posts including idealized images and descriptions of the poster's life. Compared to previous studies, the current study was designed to capture and examine the impact of a more naturalistic use of social media. Specifically, participants either browsed the feed of their own social media account versus an online newspaper (Study 1), or

the feed of a researcher-curated social media account (Study 2). This provided more realistic engagement with social media (e.g., being able to scroll through various images).

STUDY 1

Study 1 addressed the overall aims by experimentally investigating the immediate effect of viewing Facebook on body image and subsequent portion size selection in comparison to viewing a local online news website. We hypothesized that participants who spent time viewing their own Facebook account would report higher body dissatisfaction (H1) and as a result choose smaller portions (H2). Furthermore, we hypothesized body dissatisfaction would mediate the relationship between Facebook use and the choice of portion size (H3). Study 1 utilized a portion selection task as this method has been shown to be sensitive and highly predictive of what people eat and therefore the task is a suitable proxy for people's actual food intake across a diverse range of foods (Wilkinson et al., 2012).

STUDY 1 MATERIALS AND METHODS

PARTICIPANTS

Eighty-nine female Singaporean undergraduate students were initially recruited, with nine participants subsequently excluded after reporting that they did not check their Facebook account daily (pre-determined eligibility criteria). As higher rates of body dissatisfaction and disordered eating are reported in females (Griffiths et al., 2016; van den Berg et al., 2007), only female participants were recruited in Study 1. A final sample of 80 participants (age: $M = 20.75$ years, $SD = 1.75$, range = 18–25 years; BMI: $M = 21.30$, $SD = 3.23$) were randomly assigned to either browse their own Facebook account ($n = 41$) or an online local Newspaper (The Strait Times; $n = 39$). The authors note Study 1 is under powered for small effect sizes, however, post-hoc power analysis revealed we had an average of 80% power for our significant analyses. Participants were recruited with the expectation of participating in a study on social media attitudes and eating

behaviors, including internet browsing patterns, and received one research course credit for their participation.

MEASURES

Participant Characteristics. Participants were asked to provide their sex, age, nationality, and ethnicity. In addition, participants completed an adapted version of the *Facebook Use Questionnaire* (6-items; α = .81; Ross et al., 2009) as an assessment of their attitude towards the use of Facebook. The Facebook Use Questionnaire asks participants to what extent do they agree with a statement on a five-point scale from 1 (strongly disagree) to 5 (strongly agree). Participants were also asked to estimate how long they spent on social media, and specifically, Facebook, and how many friends they had on Facebook (participants could look this up if needed). Finally, weight and height were measured using a digital scale and stadiometer respectively to calculate body mass index (kg/m^2).

Hunger Assessment (Friedman et al., 1999). The HA asks participants to indicate on a 100-point visual analogue scale (VAS) their current hunger level, desire to eat, how full they are, and how much they feel they could eat at that moment. A composite hunger score was calculated (fullness item reversed; α = .96). The hunger assessment was included to control for hunger levels across groups.

Three Factor Eating Questionnaire-Revised-21 (TFEQ-R21; Cappelleri et al., 2009). The TFEQ-R21 is a self-assessment measure of three behavioral and cognitive domains of eating behavior. Cognitive restraint (CR; intentional control of food intake to maintain or reduce body weight/size; α = .80), uncontrollable eating (UE; overeating because of a lack of control over food intake; α = .83) and emotional eating (EE; consumption of more food in response to negative emotions; α = .90) and was included to describe the characteristics of the sample. Participants responded by selecting the response that best describes them on a four-point Likert type scale from 1 (definitely false) to 4 (definitely true). A single eight-point scale was re-coded as 1–4 and items 1–16 were reversed scored to calculate the mean of all items where higher scores were indicative of greater CR, UE, and EE.

Positive Affect and Negative Affect Scale (PANAS; Watson et al., 1988). The PANAS consists of ten positive (α = .90) and ten negative (α = .92) adjectives which are summed separately to reflect a participant's general affect. Participants were asked to rate, on a five-point scale from 1 (very slightly/not at all) to 5 (extremely), how they feel right now and higher scores indicate greater levels of affect.

Modified Contour Drawing Figure Rating Scale (CDFRS; Swami et al., 2010). The CDFRS is a series of different sized front-view line-drawings of female bodies from 1 (smallest) to 9 (largest). The measure has been validated across 26 countries, including Singapore by Swami and colleagues (2010). Participants were asked to select the line-drawing that best represented the following: (a) their actual body, (b) their ideal body, (c) the average body among the female undergraduate population, and (d) the average female body that would be idealized by males. To assess discrepancies across body image perceptions, (b), (c), and (d) were all subtracted from (a) respectively. Scores indicated both the magnitude and direction (i.e., scores could be positive or negative) of the body dissatisfaction. Using ideal body image as an example, negative scores represented ideal scores being greater than actual scores, therefore indicating the participant viewed their body as smaller than ideal (referred to as body dissatisfaction larger). On the other hand, positive scores represented ideal scores being smaller than actual scores, therefore indicating the participant viewed their body as larger than ideal (referred to as body dissatisfaction smaller).

Portion Selection Task. Participants completed a modified version of the portion selection task reported by Wilkinson et al. (2012). Ten foods commonly consumed in Singapore were used to ensure the task was relevant to the current sample (e.g., beef rending, fried carrot cake). Participants were asked to choose the portion size of each food item that most represented the ideal portion they would serve themselves for their next meal. Portion size increased/decreased in each image by 20 calories (ranging from 20 to 1,000 calories for each food item) and participants were able to incrementally increase/decrease the portion size by using the right and left arrow keys (see Figure 1). Calories



FIGURE 1. Example of a food item (Yang Chow Fried Rice) presented in the portion size selection task. All foods were presented in a series of 50 images that depicted incremental changes in portion sizes of the food in 20 kcal increments (ranging from 20 to 1000 kcal).

selected across the ten food items were averaged to create an overall ideal portion size (in kcal; α = .89). Participants were also asked food-related characteristics questions, for example, how much they liked each food, how filling they thought it was, how much they wanted it, how many calories that thought it contained, and how often they ate the foods.

Personal Relative Deprivation Scale (PRDS; Callan et al., 2008). The PRDS is a 4-item self-report measure to assess an individual's belief/feeling they are worse off than others by capturing their general perceptions of personal relative deprivation associated with comparing their outcomes with the outcomes of others. Participants respond by selecting the answer that best describes them on a seven-point scale from –3 (strongly disagree) to 3 (strongly agree). Two items are reverse scored, and an average score is calculated with higher scores indicating greater personal relative deprivation (α = .70).

PROCEDURE

The study was approved by the University's Institutional Review Board. On arrival, participants were handed an information sheet and informed consent was obtained. Participants completed the hunger assessment and the Three Factor Eating Questionnaire via an online survey platform (Qualtrics). Following this,

participants were randomly assigned to browse either their own personal Facebook home feed or an online local newspaper (The Strait Times) for seven minutes and were provided links to access the sites from the survey. They were verbally reminded to passively browse the websites as they normally would, with the exception that they could not post comments or messages. After seven minutes, participants were asked to return to the online survey where they were asked to list the three things that stood out most from their browsing. They then completed the Positive Affect and Negative Affect Scale and Contour Drawing Figure Rating Scale. Participants then completed the portion selection task, food-related characteristics, Perceived Relative Deprivation, demographics, and Facebook Use Questionnaire. Before leaving, participants had their height and weight measured and were debriefed. Participants also completed a Fat-Thin Implicit Association Test (Teachman & Brownell, 2001) and Crandall's Anti-Fat Attitudes Questionnaire (Crandall, 1994) during the session. These measures were included to examine whether social media exposure may influence anti-fat attitudes independent of, or in addition to body image (see supplementary materials for measure summary and results).

STATISTICAL ANALYSIS

Statistical analysis was conducted using SPSS v26 (IBM). We conducted descriptive statistics to characterize the sample and a *t*-test to analysis perceived relative deprivation. To test the effect of exposure on body image and dissatisfaction (discrepancy scores for ideal, average and attractive) we conducted a series of ANCOVAs while controlling for BMI. We adopted the *p* value of .02 ($p = .05/3$) to adjust for multiple comparisons grouping by construct (e.g., body image, body dissatisfaction, and portion size). A mediation analysis was conducted using PROCESS (v4.0) Model 4 (Hayes, 2018) entering condition 0 (control) and 1 (Facebook) as the predictor and portion size selection as the outcome with body dissatisfaction as the mediator, while controlling for BMI and baseline hunger. BMI was controlled for in analyses to examine the relationship between social media use and body dissatisfaction independent of a participant's weight and the associations between BMI and portion size.

STUDY 1 RESULTS

PARTICIPANT CHARACTERISTICS

On average participants reported they spent 1–2 hours on social media a day of which 10–30 minutes was spent on Facebook. There was a wide range of scores on the Facebook Use Questionnaire attitudes subscale (M = 18.03, SD = 4.16, range = 7–29) and the average number of Facebook friends was 535.58 (SD = 566.27, range = 40–4,000). Average scores for the Three Factor Eating Questionnaire and Perceived Relative Deprivation Scales were comparable across the two conditions (see Table 1).

BODY IMAGE

Results found a significant difference across type of exposure, whereby participants in the Facebook condition rated their actual body size as larger than the control condition, see Table 1. There was a significant main effect of condition on the perception of how much their actual size differed from ideal body size, how much their body differed from normative female size, and on the difference between actual body size and perceived male ideals, see Table 1. Each time participants in the Facebook condition had a significantly greater difference between the two scores, see Table 1. When adopting the adjusted p value, only the differences in actual body image and the discrepancy between actual and average and attractive remained significant.

PORTION SIZE SELECTION TASK

Homogeneity of variance was violated but as there is no non-parametric equivalent, an ANCOVA was conducted, controlling for BMI and baseline appetite, to examine the effect of type of exposure on portion size selection.[1] No significant difference was found between the two conditions on average portion size in kcal, see Table 1.

1 A Mann-Whitney U test also revealed no significant difference in average portion size across the two conditions (U = 2591.50, p = .575).

TABLE 1. Mean and Standard Deviations of Ratings for Outcome Variables Across Conditions.

	Facebook (n = 41)		Control (n = 39)		Sig.		
	M	*SD*	*M*	*SD*	*F*	*p*	η_p^2
Actual Body[1]^	5.49	1.49	4.41	1.48	9.36	.003*	.108
Ideal Body^	3.32	.96	3.21	.77	9.21	.969	.000
Actual-Ideal Body[2]^	2.17	1.39	1.21	1.64	5.12	.026	.062
Average Body^	3.76	1.34	3.92	.87	.20	.654	.003
Actual-Average Body[3]^	1.73	1.70	.49	1.94	6.80	.011*	.081
Male Ideal^	2.93	.82	3.03	.74	.50	.480	.007
Actual-Male Ideal[4]^	2.56	1.73	1.38	1.55	7.50	.008*	.089
Portion Size[5]^^	270.44	78.09	325.49	159.10	3.34	.071	.042
	M	*SD*	*M*	*SD*	*t*	*p*	*d*
BMI	21.87	3.35	20.70	3.03		.104	.366
Cog. Res.	15.02	3.66	14.01	3.51	1.15	.255	.281
Uncon. Eat.	20.88	4.37	21.62	4.79	.72	.474	.161
Emo. Eat.	13.59	4.21	12.74	4.08	.91	.367	.205
Rel. Dep.	3.16	1.08	2.88	.96	1.26	.212	.274

Notes. Cog. Res. = cognitive restraint; Uncon. Eat. = uncontrollable eating; Emo. Eat = emotional eating; Rel. Dep. = relative deprivation.

[1]Represents their actual body size. [2]Difference between actual body size and ideal body size. [3]Difference between actual body size and average female undergraduate body size. [4]Difference between actual body size and perceived male's ideal female body size. [5]Portion size selection in kcal. *Denotes significant to the adjusted p≤.02; ^ANCOVA controlling for BMI; ^^ANCOVA controlling for BMI and baseline hunger.

MEDIATION ANALYSIS

The direct effect (B = –29.09, SE = 27.49, p = .293) was not significant. Condition was a significant predictor of body dissatisfaction, B = .65, SE = .28, p = .025, 95% $CI_{95\%}$ [.08, 1.21] and body dissatisfaction was a significant predictor of portion size selection (b), B = –.34.56, SE = 10.75, p = .002, $CI_{95\%}$ [–55.98, –13.15]. Body dissatisfaction also significantly mediated the relationship between condition and portion size selection (B = –22.44, SE = 13.91, $CI_{95\%}$ [–54.72, –1.97]; see Figure 2). The coefficients show that relative to participants in the control condition, participants in the Facebook condition had greater body dissatisfaction, which in turn predicted smaller portion sizes.

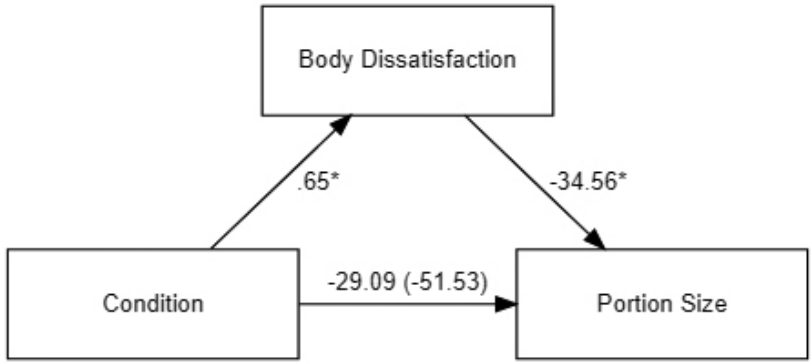


FIGURE 2. Mediation model of condition on portion size including body dissatisfaction as a mediator. Unstandardized coefficients are shown for each path and the total effect (in parentheses). *indicates significance, $p < .05$.

STUDY 1 DISCUSSION

Study 1 demonstrated that viewing one's own Facebook account rather than an online news site can result in poorer body image, largely driven by increased perceptions of one's actual body as being larger (H1 supported). As there was no difference between conditions on ideal body size, this suggests that viewing Facebook may lead to greater body dissatisfaction by making people feel fat rather than making them adopt more unrealistic ideal body ideals. Portion sizes did not significantly differ across conditions (H2 not supported); however, body dissatisfaction did significantly mediate the effect of condition on portion sizes (H3 supported). This suggests that viewing Facebook leads to increased body dissatisfaction which in turn leads to a desire to consume smaller portions. The consumption of smaller portions could be a weight management behavior to reduce calorie intake in a bid to lose weight and alleviate body dissatisfaction.

Study 1 provides initial support for a causal relationship between social media use, body dissatisfaction, and eating behavior, however it remains unknown whether there is an influence on actual food intake. One limitation to Study 1 is the small sample size. A second limitation is that we focused on one social

media platform and others need to be considered. While Facebook is the most popular social media platform generally, the use of Instagram is growing, especially among younger adults (Dixon, 2022). Given its primary focus on posting and sharing images, Instagram may lead to greater tendency to make comparisons (compared to using Facebook; (Engeln et al., 2020). Study 2 sought to address these limitations and replicate findings of Study 1.

STUDY 2

Study 2 further addressed the overall aims by experimentally examining the immediate effect of viewing one's personal Instagram account on body dissatisfaction and ad-libitum snacking. Instagram is a unique social media platform which focuses on sharing photos and videos and allows users to edit and filter their images. Notably, photo-based activities are specifically related to increased body dissatisfaction and disordered eating (Cohen et al., 2018; Meier & Gray, 2014). As previous studies of social media and body image have predominantly focused on females, Study 2 also aimed to take initial steps to examine the influence of social media on males. Initial data collection for Study 2 had previously been underway but we did not reach the desired sample size (*a-priori* power analysis) before disruptions due to COVID-19 restrictions. Given the possible changes in personal social media accounts during the global pandemic and the delay in restarting data collection, we opted to start the study afresh. We conducted preliminary analysis on the previously conducted data and pre-registered our hypotheses (https://osf.io/a8b52) for the current dataset. Study 2 was adequately powered and sought to confirm our initial findings from Study 1.

We predicted that participants who viewed their personal Instagram account (compared to control Instagram content) would report significantly increased body dissatisfaction (H1) and consume less potato chips (H2). Furthermore, females would report significantly more body dissatisfaction than males (H1a) and would consume significantly less potato chips than males (H2a). We also predicted a sex by condition interaction on body dissatisfaction (H1b), such that in the personal Instagram condition, females would report their ideal body size as smaller than their

actual whereas males will report their ideal body size as bigger than their actual. This prediction was informed by our preliminary analysis of the initial round of data collection. Likewise, we predicted a significant sex by condition interaction (H2b). Males in the personal Instagram condition would consume significantly more than males in the control Instagram condition and females in the personal Instagram condition would consume significantly less than those in the control Instagram condition. These predictions are made based on the research available that suggests males and females respond differently to media images. For example, Harrison and colleagues (2006) found reduced snack intake in females who reported high-discrepancy (mis-match between actual and ideal body image) after viewing ideal-body images (from magazines) whereas males increased snack intake. Finally, we predicted that body dissatisfaction would significantly mediate the effect of condition on snack intake (H3). Participants in the personal Instagram condition would report greater body dissatisfaction and in turn consume less potato chips.

STUDY 2 MATERIALS AND METHODS

PARTICIPANTS

The study was advertised as research investigating the relationship between social media use, everyday habits, and activities. Participants received either S$5 or one research participation credit for completion of the study. Eligibility for the study included being aged 18–65 years old,[2] having an active personal Instagram account which they check at least once a day and having no known food allergies. Power analysis using G*Power 3.1 revealed 128 participants were required to obtain a medium effect size (d = .25 with 80% power, standard alpha = .05, numerator = 1, groups = 4) for an ANCOVA with two covariates.[3] Recommendations for mediation analysis suggest a sample size of 71 for medium (.39) alpha and beta paths (Fritz & MacKinnon, 2007). Based on standardized

2 Parental consent was obtained for participants under 21 years in line with local age of contractual agreement.

3 The power analysis reported in in the pre-registration states we aimed to collect 90 males and females for this study. This is incorrect and was based on a total sample of 158.

coefficients from study 1 (α = .41, β = –.43) we consider the above sample to be sufficient for mediation analysis as well. We aimed to collect >32 participants per condition to allow for technical errors with data recording or missing data.

One hundred and seventy-one participants were initially recruited, with 23 subsequently removed for reasons such as technical/recording errors (n = 5), guessing the aims/hypotheses (n = 7), and engaging in behaviors not consistent with the instructions (n = 11). Our final sample included 81 females (age: M = 19.51 years, SD = .92, range = 18–23 years; BMI: M = 20.83, SD = 3.45) and 67 males (age: M = 22.6 years, SD = 2.03, range = 20–32 years; BMI: M = 23.82, SD = 3.80).[4] Average self-reported BMI was 22.07 (SD = 3.76) and was significantly correlated with researcher measured BMI (N = 146, r = .98, p < .001; M = 22.14, SD = 3.90).[5] Participants were randomly assigned to either browse their personal Instagram account ($n_{females}$ = 42; n_{males} = 31) or researcher created (control) Instagram account (@cool_sciencestuff; $n_{females}$ = 39; n_{males} = 36) which follows education, mathematics, and physics account profiles.

MEASURES

The following measures were collected in line with the descriptions for Study 1 (see the "Measures" section of Study 1): hunger assessment (pre: α = .90; post: α = .92), Positive Affect Negative Affect Scale (PA: α = .88; NA: α = .81) and Three Factor Eating Questionnaire (CR: α = .76; UE: α = .78; EE: α = .89). Body dissatisfaction was measured and scored as per the Contour Drawing Figure Rating Scale in the "Measures" section of Study 1 with the addition of a male version. Study 2 also included the following measures:

Manipulation Check. As it was impossible to control the images seen by the participants in the personal Instagram condition, all participants were asked to indicate what types of images they saw so a comparison could be made.

4 Assessing both Study 1 and 2 found no significant differences between samples on variables such as BMI (p = .093), age (p = .758) and baseline appetites (p = .768), suggesting relatively similar sample types.

5 Given the high correlation between self-report and researcher measured BMI, self-report BMI was used for two participants with recording errors.

Ad-libitum Snack Consumption. 60g (716.8kcal) from a packet of potato chips (FairPrice Original Flavor; 75g; 896kcal) was served to participants on a paper plate. To measure snack consumption, the plates were weighed before and after consumption with the difference in weight indicating the amount consumed in grams. After consumption, participants were asked appetite ratings of the potato chips. Specifically, participants were asked "How much did you enjoy eating the chips?" and "How delicious were the chips?" on a VAS 0 (not at all) to 100 (extremely) and these items were averaged to create a composite score, with higher scores indicating a greater preference for potato chips (α = .92). They were also asked their usual frequency of consumption of potato chips on a seven-point Likert type scale from 1 (almost never) to 7 (more than once a day).

Instagram Use. This was measured through the researchers' own questionnaire which aimed to access to what extent participants compared themselves with others on Instagram (4 items averaged; α = .82; e.g. "I compare myself to my friends when I judge how attractive I am") and their desire to look like friends and celebrities (2 items; α = .59; e.g. "I want to look like the celebrities/fitness gurus I see on my Instagram page"). General questions on use were also asked such as whether Instagram was part of their everyday activity, a preferred social media site and if they feel out of touch when they are unable to open Instagram on a scale from 1 (strongly disagree) to 5 (strongly agree; α = .84).

In addition, participants also completed the Physical Appearance Comparison Scale-3 (Schaefer et al., 2017), Compensatory Health Belief Scale (Knäuper et al., 2004), Pathogenic Weight Control Practices (Morrison et al., 2004) and Physical Activity Stages of Change Questionnaire (Marcus & Forsyth, 2009) for exploratory analyses as part of a different research project. Measure descriptions and exploratory results presented in the supplementary materials.

PROCEDURE

The study was approved by the University's Institutional Review Board. On arrival, participants were handed an information sheet and informed consent was obtained. Participants verified that

they had an Instagram account, estimated their time spent on social media, including Instagram, and were asked which gender they identify with via an online survey platform (Qualtrics). Participants indicated their current hunger and were then randomly assigned to browse either the Instagram feed of their personal account or an account curated for the study containing images of science and technology (@cool_sciencestuff). They were verbally reminded to passively browse the feeds and to not post comments. After eight minutes, participants were asked to return to the online survey where they completed the Positive Affect and Negative Affect Scale, and identify items they saw while browsing Instagram. They then completed the Contour Drawing Figure Rating Scale that matched the gender they identified at the start. Participants were then presented with potato chips and began watching a short documentary (telescope lenses; seven minutes), before they rated their hunger levels, preference for potato chips, and completed the Compensatory Health Beliefs scale, Three Factor Eating Questionnaire, social media use, and adapted Physical Appearances Comparison Scale. Finally, participants completed demographic variables such as age, dieting status, and height and weight, and asked what they felt the aim of the study was. Before leaving, participants had their height and weight measured and were debriefed.

STATISTICAL ANALYSIS

While we are unable to control what participants see in their home feed it is likely to consist of a number of different image types (Hu et al., 2014). We conducted a brief analysis of the types of images recalled by participants in the two conditions and this was also used as a manipulation check to see if participants images viewed were in line with expectations.

We conducted a series of ANCOVA's to test hypotheses 1 and 2. Firstly, we conducted a 2 (condition: personal Instagram versus control Instagram) × 2 (sex: males versus females) ANCOVA on actual and ideal body shape/size and body dissatisfaction scores (actual-ideal, actual-average, and actual-attractive) controlling for BMI. Secondly, we conducted a further 2 × 2 ANCOVA on potato chip intake (in grams) controlling for BMI and current hunger levels. For the ANCOVAs we adopted the p value of .013

($p = .05/4$) to adjust for multiple comparisons grouping by construct (e.g. body image, body dissatisfaction, intake, and participant characteristics, i.e., comparison tendency, and mood). For hypothesis 3, we conducted a meditation analysis using PROCESS (V4.0) Model 4 (Hayes, 2018). We entered condition 0 (control Instagram) or 1 (personal Instagram) as the predictor (X) and potato chip intake the outcome (Y) with body dissatisfaction as the mediator (M), while controlling for BMI and current hunger.

In addition, we conducted an ANOVA on positive and negative affect after the manipulation to check for differences in mood. Across all analyses, pairwise comparisons (Bonferroni) were conducted were appropriate. We also conducted additional exploratory analyses to help understand the results further (denoted with * in the results section).

STUDY 2 RESULTS

PARTICIPANT CHARACTERISTICS

Most participants (34.5%; $n = 51$) reported spending an average of 1–2 hours per day on social media, of which most (30.4%; $n = 45$) reported that 30–60 minutes was spent on Instagram. Participants reported average levels of frequency of comparison to others seen on Instagram ($M = 2.43$, $SD = .78$). Of those that compared themselves to others on Instagram, participants generally reported believing that they looked worse than others ($M = 3.84$, $SD = .52$) and tended to feel negative after making the comparisons ($M = 3.36$, $SD = .61$). Examining the images seen by participants in both conditions, the control Instagram condition reported viewing images related to science/education and technology ($n = 60$), with two participants reporting viewing images related to engineering and telescopes. Participants in the personal Instagram condition reported viewing a wider array of images, ranging from my friends/family ($n = 58$), celebrities ($n = 49$), fitspiration images ($n = 22$), and food ($n = 42$), which were researcher-specified items, to politics, quotes, memes, and animals, stated by participants in a free response question.

There were no significant differences in cognitive restraint and uncontrolled eating behaviors across conditions and sex (see Table 2 and supplementary materials). For emotional eating,

TABLE 2. Mean and Standard Deviations of Ratings for Outcome Variables Across Conditions.

	Personal Instagram			Control Instagram		
	Females (*n* = 42) *M* (*SD*)	**Males (*n* = 31) *M* (*SD*)**	**Total (*n* = 73) *M* (*SD*)**	**Females (*n* = 39) *M* (*SD*)**	**Males (*n* = 36) *M* (*SD*)**	**Total (*n* = 75) M (SD)**
PA	22.76 (6.65)	22.61 (7.95)	22.70 (7.18)	19.13 (5.89)	22.64 (7.58)	20.81 (6.93)
NA	14.12 (5.46)	13.23 (3.51)	13.74 (4.73)	13.21 (3.55)	13.00 (2.39)	13.11 (3.03)
CR	2.50 (.59)	2.32 (.63)	2.42 (.61)	2.49 (.51)	2.45 (.56)	2.47 (.53)
EE	2.23 (.67)	1.76 (.60)	2.03 (.68)	2.36 (.67)	2.08 (.73)	2.23 (.71)
UE	2.31 (.48)	2.15 (.51)	2.25 (.49)	2.35 (.40)	2.28 (.53)	2.32 (.47)
BMI	20.75 (3.46)	24.31 (3.46)	22.26 (3.86)	20.92 (3.47)	23.41 (4.08)	22.11 (3.95)
Actual	4.81 (1.53)	5.87 (1.34)	5.26 (1.54)	5.28 (1.64)	5.86 (1.57)	5.56 (1.62)
Ideal	3.45 (.77)	5.48 (.68)	4.32 (1.25)	3.72 (.89)	5.11 (.89)	4.39 (1.13)
Actual-Ideal	1.36 (1.56)	.39 (1.26)	.95 (1.51)	1.56 (1.48)	.75 (1.42)	1.17 (1.50)
Average	3.90 (1.10)	5.13 (1.20)	4.42 (1.29)	4.15 (.84)	5.19 (1.17)	4.65 (1.13)
Actual-Average	.90 (1.95)	.74 (1.55)	.84 (1.78)	1.13 (1.82)	.67 (1.90)	.91 (1.86)
Attractive	3.43 (.91)	5.00 (1.18)	4.09 (1.29)	3.62 (.96)	4.56 (1.03)	4.07 (1.09)
Actual-Attractive	1.38 (1.87)	.87 (1.80)	1.16 (1.85)	1.67 (1.74)	1.31 (1.74)	1.49 (1.73)
Potato Chip Intake (g)	14.36 (10.69)	18.13 (14.49)	15.96 (12.49)	15.92 (11.18)	18.42 (14.95)	17.12 (13.09)

Notes. CR = cognitive restraint (TFEQ); EE = emotional eating (TFEQ); UE = uncontrolled eating (TFEQ); BMI = body mass index.

scores were significantly higher in females ($p = .001$) as well as in the control Instagram condition ($p = .048$), but this did not remain significant when applying the adjusted p value (see Table 2 and supplementary materials).

AFFECT

There were no significant results for positive affect: sex, $F(1,144) = 2.11$, $p = .148$, condition, $F(1,144) = 2.43$, $p = .121$, sex by condition, $F(1,144) = 2.50$, $p = .116$, or negative affect: sex, $F(1,144) = .698$, $p = .405$, condition, $F(1,144) = .752$, $p = .387$, sex by condition, $F(1,144) = .27$, $p = .601$ (see Table 2).

BODY IMAGE

All means and standard deviations are presented in Table 2. Results for body dissatisfaction as calculated by actual-average and actual-attraction are presented in full in the supplementary materials.

Actual: There was no significant main effect of sex, $F(1,143) = .91$, $p = .343$, condition, $F(1,143) = 4.70$, $p = .032$; adjusted p value applied, or interaction effect, $F(1,143) = .17$, $p = .685$.

Ideal:* There was a significant main effect of sex, $F(1,143) = 116.72$, $p < .001$, $\eta_p^2 = .449$, with males reporting a significantly larger body size as being ideal compared to females. There was no significant main effect of condition, $F(1,143) = .06$, $p = .815$, or interaction effect, $F(1,143) = 4.80$, $p = .030$; adjusted p value applied.

Actual-Ideal: There was a significant main effect of sex, $F(1,143) = 74.48$, $p < .001$, $\eta_p^2 = .342$, with females reporting significantly higher body image dissatisfaction scores compared to males. There no significant main effect of condition, $F(1,143) = 4.46$, $p = .036$; adjusted p value applied, or interaction effect of sex and condition, $F(1,143) = 1.47$, $p = .227$.

AD-LIBITUM SNACK INTAKE

There was no significant main effect of sex, $F(1,142) = 1.51$, $p = .222$, or condition, $F(1,142) = .536$, $p = .465$, nor a significant interaction effect, $F(1, 142) = .004$, $p = .952$.

We conducted a second ANCOVA* that also controlled for average liking and current dietary status. There was a significant main effect of sex, $F(1,139) = 5.56$, $p = .020$, $\eta_p^2 = .08$, with males ($M = 18.97$, $SE = 1.35$) consuming significantly larger portions than females ($M = 14.50$, $SE = 1.21$). There was no significant main effect of condition, $F(1,139) = .56$, $p = .454$, or interaction effect, $F(1,139) = .01$, $p = .942$.

MEDIATION ANALYSIS

There was no significant direct effect of condition on potato chip intake controlling for body dissatisfaction ($B = –1.94$, $SE = 1.93$,

$CI_{95\%}$ [–5.76, 1.88]), or indirect effect (B = .32, SE = .40, $CI_{95\%}$ [–.24, 1.26]); see Figure 3.

EXPLORATORY ANALYSES

We conducted a series of exploratory analyses that examined the moderating effect of body image and dissatisfaction on potato chip intake.[6] We conducted separate models for males and females (due to sex differences in variables such as ideal body image and potato chip intake) using PROCESS Model 1. Condition was entered as the predictor (X) and potato chip intake as the outcome (Y). Body image (actual and ideal) and body dissatisfaction (actual-ideal, actual-average, actual-attractive; shown in the supplementary materials) were entered in turn as moderators (W). Hunger, BMI, average liking of chips and current dieting status were entered as covariates.

Females. For actual body image scores, the overall model was significant, $F(7,73) = 6.88$, $p < .001$, and explained 39.75% of the variance, but the interaction was not significant, $R^2_{change} = .00$, $F(1,73) = .50$, $p = .480$. For ideal body image scores, the overall model was significant, $F(7,73) = 8.43$, $p < .001$, and explained 44.71% of the variance. The interaction was also significant, $R^2_{change} = .05$, $F(1,73) = 6.39$, $p = .014$; see Figure 4. Specifically, female in the personal Instagram condition consumed significantly less potato chips than those in the control Instagram condition when they reported a lower body image score as ideal (i.e., a thinner body was rated as ideal), $B = –6.33$, $SE = 2.82$, $t(73) = –2.25$, $p = .028$ [$CI_{95\%} = –11.94$ –.72]. There was no significant difference in intake when participants reported an average ($B = –1.12$, $SE = 1.91$, $t(73) = –.58$, $p = .561$ [$CI_{95\%} = –4.93$–2.69]) or higher ($B = 4.10$, $SE = 2.81$, $t(73) = 1.46$, $p = .149$ [$CI_{95\%} = –1.50$–9.70]) body image scores as ideal between the two conditions. Results for body dissatisfaction variables are presented in the supplementary materials.

Males. For actual body image scores, the overall model was significant, $F(7,58) = 6.57$, $p < .001$, and explained 44.24% of

6 Power analysis for moderation: R^2 increase requires 55 participants to detect medium effect at 80% ($p = .05$) for 1 tested predictor and 7 total predictors.

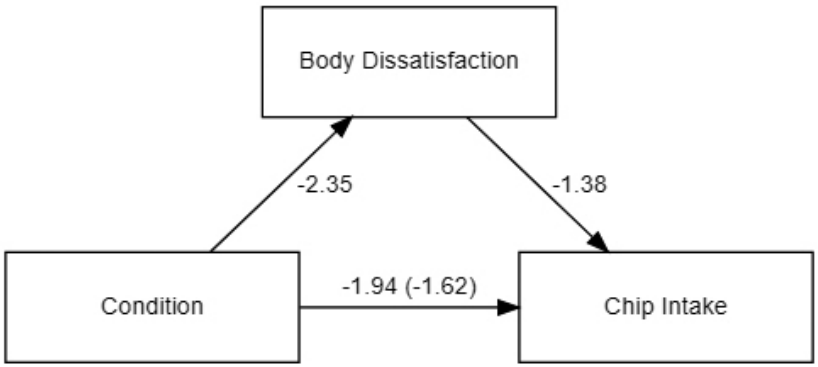


FIGURE 3. Mediation model of condition on chip intake including body dissatisfaction as a mediator. Unstandardized coefficients are shown for each path and the total effect (in parentheses).

the variance. The interaction was also significant, R^2_{change} = .06, $F(1,58)$ = 5.73, p = .020; see Figure 5. Specifically, males in the personal Instagram condition consumed significantly less potato chips than those in the control Instagram condition when they reported their actual body image as being lower (i.e., thinner), B = –8.95, SE = 4.20, $t(58)$ = –2.13, p = .037 [$CI_{95\%}$ = –17.36–.54]. When participants reported an average (B = –1.82, SE = 2.94, $t(58)$ = –.62, p = .540[$CI_{95\%}$ = –7.71–4.08]) or higher (B = 5.32, SE = 4.18, $t(58)$ = 1.27, p = .208 [$CI_{95\%}$ = –3.05–13.69]) body image score, intake across the conditions was similar. Results for ideal body image and body dissatisfaction variables are presented in the supplementary materials.

For ideal body image scores, the overall model was significant, $F(7,58)$ = 5.72, p < .001, and explained 40.84% of the variance, but the interaction was not significant, R^2_{change} = .03, $F(1,58)$ = 2.68, p = .107. Results for body dissatisfaction variables are presented in the supplementary materials.

STUDY 2 DISCUSSION

Results from Study 2 supported some of our hypotheses. In terms of body image and body dissatisfaction, females reported

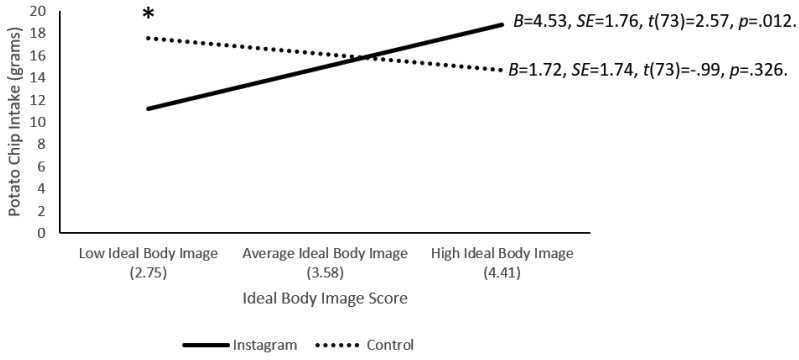


FIGURE 4. Moderation of the effect of condition on kcal intake by ideal body image for females (M, −/+ 1SD, *denotes significant difference, $p < .05$).

significantly greater body dissatisfaction than males for all levels of discrepancy (actual-ideal, average and attractive; H1a). We also found that actual body image was reported as significantly larger by participants that viewed their personal Instagram compared to the control (H1), but this did not remain significant when applying the p value adjustment. We did not find support for significant sex by condition interactions (H1b). In terms of potato chip intake, we only found support for males consuming significantly more than females and found our proposed mediation model was not significant (H3). There was also no significant sex by condition interaction, where participants from the personal Instagram condition did not consume significantly more potato chips than their same gender counterparts in the control condition (H2b).

Exploratory moderation analyses revealed that ideal body image for females and actual body image for males moderated the effect of condition on potato chip intake. Females in the personal Instagram condition consumed significantly less potato chips when they reported a smaller body size as being ideal compared to females in the control Instagram condition. In contrast, males in the personal Instagram condition consumed significantly less potato chips when they selected a smaller body

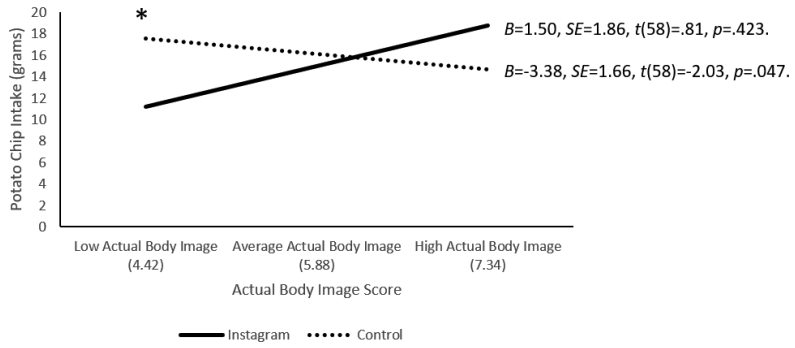


FIGURE 5. Moderation of the effect of condition on kcal intake by actual body image for males (M, –/+ 1SD, *denotes significant difference, $p < .05$).

size as representing their actual body compared to males in the control Instagram condition.

DISCUSSION

Our first overall aim was to examine the impact of social media on body image (aim 1). We found some support that brief exposure (less than 10 minutes) has a negative effect on body image. Specifically, female participants in the social media condition in Study 1 reported their actual body image as significantly larger than those in the control condition (online news website), but this was not replicated in Study 2 (after p value adjustment). We also found that after viewing their personal Instagram (Study 2), females reported a significantly smaller ideal body image whereas males reported a significantly larger ideal body image. We found support for our hypothesis that social media exposure would result in greater body dissatisfaction (H1) in Study 1, however, this was not the case in Study 2. Participants who viewed their own social media in Study 2 did not report significantly more body dissatisfaction than those who viewed the control account. It is likely that this is due to the condition

versus sex interaction. Although the interactions were not significant overall, there were pairwise comparisons in which the effect of social media resulted in females being more likely to prefer a smaller body compared to males who were more likely to prefer a larger body. A strength of the current studies was the use of participants' own social media accounts, and therefore, engagement with the types of content they encounter in daily life (i.e., based on who they follow,'like,' and who they are friends with). This is in comparison to previous studies where exposure to social media involved experimenter-curated 'ideal' images that have be specifically selected (e.g., Fardouly & Holland, 2018; Tiggemann & Anderberg, 2020). We show that the content participants view, and the images they choose to view daily, even for a short period of time, can have a detrimental impact on their body image. These findings are supported across social media platforms, and, although mechanisms may differ, across both males and females.

Our second aim was to assess the impact of social media on eating behavior (aim 2). We found no significant effect of a short bout of social media use on eating behavior (H2 unsupported). However, this prediction was informed by the current available literature which is predominantly based on correlational findings, in which a causal relationship between patterns of social media use and restrained/disordered eating behaviors are unclear (e.g., Bennett et al., 2020; Griffiths et al., 2018; McLean et al., 2015; Modica, 2020; Saunders & Eaton, 2018; Stronge et al., 2015). Our studies provide initial tests of the causal relationship between social media and eating behaviors, but further work is needed measuring other types of food-related behaviors.

Finally, we aimed to investigate the relationship between social media, body image, and eating behaviors (aim 3). We found that body dissatisfaction mediated the relationship between condition and portion size selection in study 1 (H3). Here, participants who viewed Facebook, relative to an online news website (control), reported increased body dissatisfaction which in turn led to selection of smaller portions. The consumption of smaller portions could be a weight management behavior to reduce calorie intake in a bid to lose weight and alleviate body dissatisfaction. We did not find a similar pattern of results when testing the mediation model in Study 2. The relationship between social media, body image, and eating behaviors has been summarized

as complex and inconsistent in a systematic review, and there may be other moderators and boundary conditions contributing to their relationship (Rounsefell et al., 2020). Exploratory analysis of Study 2 did suggest that gender differences may play a role in how social media effects body dissatisfaction and in turn eating behaviors.

Specifically, in our exploratory moderation analyses we found significant interactions between condition and body image perceptions for females and males. When females viewed their personal Instagram and rated a thinner body as ideal, they consumed significantly less potato chips than participants in the control Instagram condition. It is possible that viewing social media acts as a catalyst to individuals reducing calorie intake when they rate a smaller body as being ideal. In contrast, for males it was when they reported their actual body size as being thinner that they consumed significantly less potato chips when in the personal Instagram condition compared to the control Instagram condition. These findings suggest that the typical tendency for males who perceive their bodies as smaller to consume more potato chips as a snack (i.e., in the control condition) may be suppressed following exposure to their personal social media content. This could reflect a motivation to maintain their current body size following comparisons to others on Instagram. Social media appears to influence males and females differently in terms of both body image and eating behaviors. However, these are preliminary findings and further work is required to tease out the exact mechanisms and to test different foods and eating behavior environments.

Strengths of our study include participants viewing their own social media and the use of multimethod measurements of eating behaviors. Nevertheless, it is important to highlight the limitations of examining eating behaviors in research. We utilized a portion size selection task as an analogue for desired food intake (Study 1). While research suggests portion size tasks are a strong indicator of desired food choice, they do not measure food intake and therefore caution is needed interpretating findings. In addition, we used an ad-libitum snack task over a small period with just one food while watching a neutral documentary (Study 2). A wider range of test food, longer consumption periods and checks of boredom eating (Domoff et al., 2014; Koball et al., 2012) could have resulted in different findings. Future studies should look

to capture food intake across a longer period and a variety of foods, such as whether viewing social media influences immediate versus delayed food intake. In line with this, while we assessed, and controlled for, current hunger levels in our analyses, the study was completed by participants across the day and therefore participants may have stronger/weaker desires to consume snacks. Another limitation is that we only collected post-task measures of body image and body dissatisfaction. This was to prevent participant burden and participants were randomly assigned to the experimental conditions to reduce influences from individual differences. Finally, we are unable to know the exact images and content that participants were viewing on their social media accounts, and consequently, are unable to control/standardize the images and content they were exposed to. However, our overall design and the use of participants viewing their own social media provides a more ecologically valid manipulation compared to researcher curated images, since participants can 'scroll through' content of their real accounts. That said, we instructed our participants to be passive users (monitoring social media without direct engagement) of social media compared to active users (direct engagement with others on social media) and further research on the impact of type of engagement is required (Trifiro & Gerson, 2019; Valkenburg et ak., 2022).

CONCLUSION

In conclusion, our studies suggest brief exposure to social media may lead to poorer perceptions of one's body but may not influence subsequent eating behaviors. However, we saw initial support for restricted intake in males and females who perceived their own body as smaller and wanted a smaller body, respectively. Our results have significant implications on both practice and contributing to the knowledge available on social media and body dissatisfaction. Coming on the heels of the release of the Facebook files (Wells et al., 2021), this paper further adds affirmation to the wealth of literature on social media and body image, while taking the initial steps to examine the causal relationship between social media, body image, and eating behavior. We also take initial steps in differentiating the impact and subsequent behaviors of social media between males and females.

Additional work is required here to further tease apart sex and individual differences.

REFERENCES

Bennett, B. L., Whisenhunt, B. L., Hudson, D. L., Wagner, A. F., Latner, J. D., Stefano, E. C., & Beauchamp, M. T. (2020). Examining the impact of social media on mood and body dissatisfaction using ecological momentary assessment. *Journal of American College Health*, *68*(5), 502–508. https://doi.org/10.1080/07448481.2019.1583236

Callan, M. J., Ellard, J., & Hodgins, D. (2008). Gambling as a search for justice: Examining the role of personal relative deprivation in gambling urges and gambling behavior . *Article in Personality and Social Psychology Bulletin*, *34*(11), 1514–1529. https://doi.org/10.1177/0146167208322956

Cappelleri, J. C., Bushmakin, A. G., Gerber, R. A., Leidy, N. K., Sexton, C. C., Lowe, M. R., & Karlsson, J. (2009). Psychometric analysis of the Three-Factor Eating Questionnaire-R21: Results from a large diverse sample of obese and non-obese participants. *International Journal of Obesity*, *33*, 611–620. https://doi.org/10.1038/ijo.2009.74

Cheon, B. K., & Hong, Y. Y. (2017). Mere experience of low subjective socioeconomic status stimulates appetite and food intake. *Proceedings of the National Academy of Sciences of the United States of America*, *114*(1), 72–77. https://doi.org/10.1073/pnas.1607330114

Cheon, B. K., Lim, E. X., McCrickerd, K., Zaihan, D., & Forde, C. G. (2018). Subjective socioeconomic status modulates perceptual discrimination between beverages with different energy densities. *Food Quality and Preference*, *68*, 258–266. https://doi.org/10.1016/j.foodqual.2018.03.010

Cohen, R., Newton-John, T., & Slater, A. (2018). 'Selfie'-objectification: The role of selfies in self-objectification and disordered eating in young women. *Computers in Human Behavior*, *79*, 68–74. https://doi.org/10.1016/j.chb.2017.10.027

Crandall, C. S. (1994). Prejudice against fat people: Ideology and self-interest. *Journal of Personality and Social Psychology*, *66*(5), 882–894.

Dixon, S. (2022). Instagram: Distribution of global audiences 2022, by age group. https://www.statista.com/statistics/325587/instagram-global-age-group/

Domoff, S. E., Meers, M. R., Koball, A. M., & Musher-Eizenman, D. R. (2014, December 12). The validity of the Dutch Eating Behavior Questionnaire: Some critical remarks. *Eating and Weight Disorders*, (Vol. 19, pp. 137–144). Springer. https://doi.org/10.1007/s40519-013-0087-y

Engeln, R., Loach, R., Imundo, M. N., & Zola, A. (2020). Compared to Facebook, Instagram use causes more appearance comparison and lower body satisfaction in college women. *Body Image*, *34*, 38–45. https://doi.org/10.1016/j.bodyim.2020.04.007

Fardouly, J., & Holland, E. (2018). Social media is not real life: The effect of attaching disclaimer-type labels to idealized social media images on women's

body image and mood. *New Media and Society*, *20*(11), 4311–4328. https://doi.org/10.1177/1461444818771083

Fardouly, J., Pinkus, R. T., & Vartanian, L. R. (2017). The impact of appearance comparisons made through social media, traditional media, and in person in women's everyday lives. *Body Image*, *20*, 31–39. https://doi.org/10.1016/j.bodyim.2016.11.002

Fardouly, J., & Vartanian, L. R. (2016). Social media and body image concerns: Current research and future directions. *Current Opinion in Psychology*, *9*, 1–5. Elsevier. https://doi.org/10.1016/j.copsyc.2015.09.005

Friedman, M. I., Ulrich, P., & Mattes, R. D. (1999). A figurative measure of subjective hunger sensations. *Appetite*, *32*(3), 395–404. https://doi.org/10.1006/appe.1999.0230

Fritz, M. S., & MacKinnon, D. P. (2007). Required sample size to detect the mediated effect. *Psychological science*, *18*(3), 233–239.

Gibbons, F. X. (1986). Social comparison and depression: Company's effect on misery.: EBSCOhost. *Journal of Personality and Social Psychology*, *51*(1), 140–148.

Gomez, M., Klare, D., Ceballos, N., Dailey, S., Kaiser, S., & Howard, K. (2021). *Do You Dare to Compare?: The Key Characteristics of Social Media Users Who Frequently Make Online Upward Social Comparisons*. https://doi.org/10.1080/10447318.2021.1976510

Griffiths, S., Hay, P., Mitchison, D., Mond, J. M., McLean, S. A., Rodgers, B., Massey, R., & Paxton, S. J. (2016). Sex differences in the relationships between body dissatisfaction, quality of life and psychological distress. *Australian and New Zealand Journal of Public Health*, *40*(6), 518–522. https://doi.org/10.1111/1753-6405.12538

Griffiths, S., Murray, S. B., Krug, I., & McLean, S. A. (2018). The contribution of social media to body dissatisfaction, eating disorder symptoms, and anabolic steroid use among sexual minority men. *Cyberpsychology, Behavior, and Social Networking*, *21*(3), 149–156. https://doi.org/10.1089/cyber.2017.0375

Harrison, K., Taylor, L. D., & Marske, A. L. (2006). Women's and men's eating behavior following exposure to ideal-body images and text. *Communication Research*, *33*, 507–529. https://doi.org/10.1177/0093650206293247

Hawkins, L. K., Farrow, C., & Thomas, J. M. (2020). Do perceived norms of social media users' eating habits and preferences predict our own food consumption and BMI? *Appetite*, *149*, 104611. https://doi.org/10.1016/j.appet.2020.104611

Hayes, A. F. (2018). *Introduction to mediation, moderation, and conditional process analysis* (2nd ed.). Guildford Press.

Hogue, J. V., & Mills, J. S. (2019). The effects of active social media engagement with peers on body image in young women. *Body Image*, *28*, 1–5. https://doi.org/10.1016/j.bodyim.2018.11.002

Holland, G., & Tiggemann, M. (2016). A systematic review of the impact of the use of social networking sites on body image and disordered eating outcomes. *Body Image*, *17*, 100–110. Elsevier Ltd. https://doi.org/10.1016/j.bodyim.2016.02.008

Hu, Y., Manikonda, L., & Kambhampati, S. (2014). What We Instagram: A First Analysis of Instagram Photo Content and User Types. *Proceedings of the International AAAI Conference on Web and Social Media*, *8*(1).

Hunt, M. G., Marx, R., Lipson, C., & Young, J. (2018). No more FOMO: Limiting social media decreases loneliness and depression. *Journal of Social and Clinical Psychology*, *37*(10), 751–768. https://doi.orgh/10.1521/jscp.2018.37.10.751

Ioannidis, K., Taylor, C., Holt, L., Brown, K., Lochner, C., Fineberg, N. A., Corazza, O., Chamberlain, S. R, Roman-Urrestarazu, A., & Czabanowska, K. (2021). Problematic usage of the internet and eating disorder and related psychopathology: A multifaceted, systematic review and meta-analysis. *Neuroscience & Biobehavioral Reviews*, *125*, 569–581. https://doi.org/10.1016/J.NEUBIOREV.2021.03.005

Knäuper, B., Rabiau, M., Cohen, O., & Patriciu, N. (2004). Compensatory health beliefs: Scale development and psychometric properties. *Psychology and Health*, *19*(5), 607–624. https://doi.org/10.1080/0887044042000196737

Koball, A. M., Meers, M. R., Storfer-Isser, A., Domoff, S. E., & Musher-Eizenman, D. R. (2012). Eating when bored: Revision of the emotional eating scale with a focus on boredom. *Health Psychology*, *31*(4), 521–524. https://doi.org/10.1037/a0025893

Lee, M., & Lee, H. H. (2021). Social media photo activity, internalization, appearance comparison, and body satisfaction: The moderating role of photo-editing behavior. *Computers in Human Behavior*, *114*, 106579. https://doi.org/10.1016/J.CHB.2020.106579

Livingston, J., Holland, E., & Fardouly, J. (2020). Exposing digital posing: The effect of social media self-disclaimer captions on women's body dissatisfaction, mood, and impressions of the user. *Body Image*, *32*, 150–154. https://doi.org/10.1016/J.BODYIM.2019.12.006

Marcus, B. H., & Forsyth, L. H. (2009). Motivating people to be physically active. In *Motivating People to Be Physically Active*. Human Kinetics Publishers Inc. https://doi.org/10.5040/9781492596356

McLean, S. A., Paxton, S. J., Wertheim, E. H., & Masters, J. (2015). Photoshopping the selfie: Self photo editing and photo investment are associated with body dissatisfaction in adolescent girls. *International Journal of Eating Disorders*, *48*(8), 1132–1140. https://doi.org/10.1002/eat.22449

Meier, E. P., & Gray, J. (2014). Facebook photo activity associated with body image disturbance in adolescent girls. *Cyberpsychology, Behavior, and Social Networking*, *17*(4), 199–206. https://doi.org/10.1089/cyber.2013.0305

Modica, C. A. (2020). The associations between Instagram use, selfie activities, appearance comparison, and body dissatisfaction in adult men. *Https://Home.Liebertpub.Com/Cyber*, *23*(2), 90–99. https://doi.org/10.1089/CYBER.2019.0434

Morrison, T. G., Kalin, R., & Morrison, M. A. (2004). Body-image evaluation and body-image investment among adolescents: A test of sociocultural and social comparison theories. *Adolescence*, *39*, 571–592.

Ross, C., Orr, E. S., Sisic, M., Arseneault, J. M., Simmering, M. G., & Orr, R. R. (2009). Personality and motivations associated with Facebook use. *Computers in Human Behavior*, *25*(2), 578–586. https://doi.org/10.1016/j.chb.2008.12.024

Rounsefell, K., Gibson, S., McLean, S., Blair, M., Molenaar, A., Brennan, L., Truvy, H., & McCaffrey, T. A. (2020). Social media, body image and food choices in healthy young adults: A mixed methods systematic review. *Nutrition and Dietetics*, *77*(1), 19–40. https://doi.org/10.1111/1747-0080.12581

Ryding, F. C., & Kuss, D. J. (2020). The use of social networking sites, body image dissatisfaction, and body dysmorphic disorder: A systematic review of psychological research. *Psychology of Popular Media*, *9*(4), 412–435. https://doi.org/10.1037/ppm0000264

Saiphoo, A. N., & Vahedi, Z. (2019, December). A meta-analytic review of the relationship between social media use and body image disturbance. *Computers in Human Behavior*, *101*, 259–275. Elsevier Ltd. https://doi.org/10.1016/j.chb.2019.07.028

Saunders, J. F., & Eaton, A. A. (2018). Snaps, selfies, and shares: How three popular social media platforms contribute to the sociocultural model of disordered eating among young women. *Cyberpsychology, Behavior, and Social Networking*, *21*(6), 343–354. https://doi.org/10.1089/cyber.2017.0713

Schaefer, L. M., Harriger, J. A., Heinberg, L. J., Soderberg, T., & Kevin Thompson, J. (2017). Development and validation of the sociocultural attitudes towards appearance questionnaire-4-revised (SATAQ-4R). *International Journal of Eating Disorders*, *50*(2), 104–117. https://doi.org/10.1002/eat.22590

Sim, A. Y., Lim, E. X., Forde, C. G., & Cheon, B. K. (2018). Personal relative deprivation increases self-selected portion sizes and food intake. *Appetite*, *121*, 268–274. https://doi.org/10.1016/j.appet.2017.11.100

Sim, A. Y., Lim, E. X., Leow, M. K., & Cheon, B. K. (2018). Low subjective socioeconomic status stimulates orexigenic hormone ghrelin — A randomised trial. *Psychoneuroendocrinology*, *89*, 103–112. https://doi.org/10.1016/j.psyneuen.2018.01.006

Stice, E. (2002). Risk and maintenance factors for eating pathology: A meta-analytic review. *Psychological Bulletin*, *128*(5), 825–848. https://doi.org/10.1037/0033-2909.128.5.825

Stronge, S., Greaves, L. M., Milojev, P., West-Newman, T., Barlow, F. K., & Sibley, C. G. (2015). Facebook is Linked to Body Dissatisfaction: Comparing Users and Non-Users. *Sex Roles*, *73*(5–6), 200–213. https://doi.org/10.1007/s11199-015-0517-6

Swami, V., Frederick, D. A., Aavik, T., Alcalay, L., Allik, J., Anderson, D., Andrianto, S., Arora, A., Brännström, A., Cunningham, J., Danel, D., Doroszewicz, K., Forbes, G. B., Furnham, A., Greven, C. U., Halberstadt, J., Hao, S., Haubner, T., Hwang, C. S., Inman, M., . . . Zivcic-Becirevic, I. (2010). The attractive female body weight and female body dissatisfaction in 26 countries across 10 world regions: Results of the international body project I. *Personality and Social Psychology Bulletin*, *36*(3), 309–325. https://doi.org/10.1177/0146167209359702

Teachman, B. A., & Brownell, K. D. (2001). Implicit anti-fat bias among health professionals: Is anyone immune? *International Journal of Obesity*, *25*(10), 1525–1531. https://doi.org/10.1038/sj.ijo.0801745

Tiggemann, M., & Anderberg, I. (2020). Social media is not real: The effect of 'Instagram vs reality' images on women's social comparison and body image. *New Media and Society*, *22*(12), 2183–2199. https://doi.org/10.1177/1461444819888720

Tiggemann, M., Hayden, S., Brown, Z., & Veldhuis, J. (2018). The effect of Instagram "likes" on women's social comparison and body dissatisfaction. *Body Image*, *26*, 90–97. https://doi.org/10.1016/j.bodyim.2018.07.002

Trifiro, B. M., & Gerson, J. (2019). Social media usage patterns: Research note regarding the lack of universal validated measures for active and passive use. *Social Media + Society*. https://doi.org/10.1177/2056305119848743

Valkenburg, P. M., van Driel, I. I., & Beyens, I. (2022). The associations of active and passive social media use with well-being: A critical scoping review. *New Media & Society*, *24*(2), 530–549. https://doi.org/10.1177/14614448211065425

van den Berg, P., Paxton, S. J., Keery, H., Wall, M., Guo, J., & Neumark-Sztainer, D. (2007). Body dissatisfaction and body comparison with media images in males and females. *Body Image*, *4*(3), 257–268. https://doi.org/10.1016/j.bodyim.2007.04.003

Vaterlaus, J. M., Patten, E. V., Roche, C., & Young, J. A. (2015). #Gettinghealthy: The perceived influence of social media on young adult health behaviors. *Computers in Human Behavior*, *45*, 151–157. https://doi.org/10.1016/J.CHB.2014.12.013

Watson, D., Clark, L. A., & Tellegen, A. (1988). Development and validation of brief measures of positive and negative affect: The PANAS scales. *Journal of Personality and Social Psychology*, *54*(6), 1063–1070. https://doi.org/10.1037//0022-3514.54.6.1063

Wells, G., Horwitz, J., & Seetharaman, D. (2021). Facebook Knows Instagram Is Toxic for Teen Girls, Company Documents Show - WSJ. Retrieved October 5, 2021, from The Wall Street Journal website: https://www.wsj.com/articles/facebook-knows-instagram-is-toxic-for-teen-girls-company-documents-show-11631620739?mod=hp_lead_pos7&mod=article_inline

Wilkinson, L. L., Hinton, E. C., Fay, S. H., Ferriday, D., Rogers, P. J., & Brunstrom, J. M. (2012). Computer-based assessments of expected satiety predict behavioural measures of portion-size selection and food intake. *Appetite*, *59*(3), 933–938. https://doi.org/10.1016/j.appet.2012.09.007